

Element C

Connection to Prior Research

Arthritis affects millions of people. Many products dedicated to relieve symptoms of arthritis implement similar methods of relief and restoration, those including: heating, detoxification, external assistance. The quality and effectiveness of the aspects vary wildly across products, some are more specialized than others or have more methods for relief yet less effective overall. Researching the said products reveals how price range influences quality and effectiveness, for example an exoskeleton glove would be the most effective but it is not suited for daily life and is often too expensive. Other products can be cheaper and worn for daily leisure but lack the quality of treatment for arthritis. In addition, other aspects of our project are intended to address aspects that many products sometimes fail to address themselves, for example customization, customer safety, and easy accessibility. By defining the constraints and requirements of the intended project based on existing products with their own respective objectives and solutions, our group is able to keep this project plausible within manufacturing based on current timeline and available resources.

Requirements List

High Priority Requirements:

1. The glove will ensure customer safety by regulating temperature, including failsafes to mitigate abnormal temperature fluctuations, and protect from voltage surges by a measure of internal pass/fail tests, after running the product at maximum intensity. Tests will pass if no injuries or irritations occur to the user and positive feedback is received.
2. The glove will assist grip, measured by stakeholder pain relief and ease in grip using a rating scale of 1 - 10 (1 being the worst), measured before and after using the glove, intending for a difference of 3 or greater. General objects that patients have difficulty holding/using will be identified for testing.
3. The glove will heat the hand(s) up to a temperature of 104 degrees Fahrenheit, not exceeding 111 degrees Fahrenheit, measured in degrees Fahrenheit using a built-in thermometer, heating for 15-20 minutes.
4. Built-in glove sensors will sense/perceive changes in temperature, grip setting, movement, and vibration in the glove worn on the hand afflicted by arthritis, measured using data from built-in sensors, including, but not limited to flex sensors, thermometers, timers, etc. Valid data signifies that the product test has passed, with invalid data implying the product is flawed.
5. The glove shall easily release the grip for people with arthritis measured with a consumer rating of at least a score of 7 or greater, measured by stakeholder feedback on a scale of 1 to 10, 1 being the worst.

6. The glove will manipulate the soft tissue in the hands for people with hand arthritis for 15 minute sessions with a built in timer to measure the duration for each session. Timer data signifies whether sessions were 15 minutes or not.

Requirements:

1. The glove will adapt to basic app settings for people with hand arthritis by continuously updating to any recent changes made on the connected app via bluetooth measured by the valid connection of the app to the product and the ability for stakeholders (consumers) to manipulate and customize determined aspects of the product.

Project Update 11/19/25

After receiving feedback from Drexel professors and various peers, we have decided to number and prioritize requirements in the order we will focus on. Our main feature is the assistive grip, so for the time being it will be the main development onto the prototype over all other priorities. Then, we will integrate the heating, and finally the vibration. The app will be only available if we have time. The safety of the product will still continue to be viewed as a top priority during prototype development.